CHAPTER FIVE

Conservation Status of Killer Whales, *Orcinus orca*, in the Strait of Gibraltar

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Abstract

Killer whales (*Orcinus orca*) in the Mediterranean Sea are currently restricted to the Strait of Gibraltar and surrounding waters. Thirty-nine individuals were present in 2011, with a well-differentiated social structure, organized into five pods. Killer whale occurrence in the Strait is apparently related to the migration of their main prey, Atlantic bluefin tuna

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(*Thunnus thynnus*). In spring, whale distribution was restricted to shallow waters off the western coast of the Strait where all pods were observed actively hunting tuna. In summer, the whales were observed in the shallow central waters of the Strait. A relatively new feeding strategy has been observed among two of the five pods. These two pods interact with an artisanal drop-line fishery. Pods depredating the fishery had access to larger tuna in comparison with pods that were actively hunting. The Strait of Gibraltar killer whales are socially and ecologically different from individuals in the Canary Islands. Molecular genetic research has indicated that there is little or no female-mediated gene migration between these areas. Conservation threats include small population size, prey depletion, vessel traffic, and contaminants. We propose the declaration of the Strait of Gibraltar killer whales as an endangered subpopulation. A conservation plan to protect the Strait of Gibraltar killer whales is urgently needed, and we recommend implementation of a seasonal management area where activities producing underwater noise are restricted, and the promotion of bluefin tuna conservation.

1. INTRODUCTION

Ancient naturalists described marine creatures in the Mediterranean Sea, but the identification of these creatures as marine mammals was uncertain because they were believed to be sea monsters. Moreover, the descriptions were usually symbolic and inclusive of mythology. Around the first century, Pliny the Elder produced what is likely the earliest published account of killer whales (*Orcinus orca*) (Notarbartolo di Sciara, 1987). In *Naturalis Historia* (Book IX.5), Pliny described whales that were the enemies of large whales because they fed on pregnant females and calves in the Gulf of Cadiz. These were likely to have been killer whales. In mid-winter, large whales were apparently abundant in the waters off Cadiz, but in summer they sought refuge in sheltered coastal areas to give birth, and it was there that killer whales attacked them. In the second century, Strabo, in *Geografia* (Book II 2,7), described Turdetania, a region of the southern Iberian Peninsula, and included what were likely killer whales among the marine life. Strabo referred to these whales as '*Oryges*' (Latin for gazelle or goat). In the same century, Aelianus, in *De Natura Animalium* (Book XV. 2), noted the presence of what were probably killer whales in the Strait of Bonifacio, between Corsica and Sardinia in the Mediterranean Sea. Aelianus referred to these whales as '*Marine Arietes*' (Latin for marine ram), and described their hunting techniques, noting that they were capable of 'intentional stranding' to penetrate the caves where seals were resting. These seals were most likely monk seals (*Monachus monachus*), which were abundant and widespread throughout the Mediterranean Sea, the North-African Atlantic coast, and

Macaronesian islands in that era (Brasseur et al., 1997; Israëls, 1992; King, 1956; Ronald and Healey, 1978).

In the 16th century, Horozco (1598) reported that tuna (Atlantic bluefin tuna, *Thunnus thynnus*) entered the Santi Pretri River (Cadiz, southern Spain), escaping from killer whales. Killer whales were reported to be common along the Atlantic coast of the Iberian Peninsula (Graells, 1889), and more particularly in the Gulf of Cadiz (Cornide, 1788; De Huerta, 1624), where it was said to be the most common marine mammal (Machado y Nuñez, 1869). Nevertheless, historically, the species was considered to be rare or absent in the Mediterranean Sea (Borrell et al., 2000; Cabrera, 1914; Casinos and Vericad, 1976; Cornide, 1788; Cuvier, 1836; Di Natale and Mangano, 1983; Duguy and Cyrus, 1973; Duguy et al., 1983; Graells, 1889; Graus et al., 1986; Machado y Nuñez, 1869; Notarbartolo di Sciara, 1987; Raga et al., 1985; Van Beneden, 1888).

Potential food resources for killer whales in the Mediterranean Sea were depleted throughout history by human activities such as the massive hunting of monk seals (Brasseur et al., 1997; Sal'nikov, 1959) and fin whales (*Balaenoptera physalus*). Hunting led to the near complete disappearance of the fin whale Mediterranean subpopulation south-west of the Strait of Gibraltar in the Atlantic Ocean (Aguilar, 1985, 2006; Clapham et al., 2008; Sanpera and Aguilar, 1992). It has been suggested that the decline of killer whales in the Atlantic Ocean may have been due to a shortage of prey, such as humpback (*Megaptera novaeangliae*) and other baleen whales (Duguy and Robineau, 1973). It is possible that the current killer whale diet, which is based on bluefin tuna (García-Tiscar, 2009), may be the result of adaptation in response the decline of the marine mammal populations that were apparently a part of their historical diet. However, another hypothesis is that two killer whale ecotypes (one specializing in predation of marine mammals and the other predating fish) inhabited the Mediterranean and nearby areas of the Atlantic historically (Esteban, 2015). The coexistence of these two killer whale ecotypes has been described in other areas such as the North Atlantic (Foote et al., 2009), and the Pacific Ocean (Ford et al., 1998).

From 1999 to 2011, we conducted research to investigate the distribution, ecology, social structure, population structure, and demography of killer whales in the Strait of Gibraltar. Here we have reviewed research on the Strait of Gibraltar killer whales and include a discussion of conservation threats, management actions, and current conservation. We conclude with recommendations regarding measures for the future protection of the Strait of Gibraltar killer whales.

2. KILLER WHALE DISTRIBUTION

Killer whales have a wide distribution throughout all oceans and seas, from polar waters to the equator (Forney and Wade, 2007; Heyning and Dahlheim, 1988; Leatherwood and Dalheim, 1978). They are known to be common in many coastal areas, particularly at high latitudes, but they are also present in high seas and tropical waters (Forney and Wade, 2007; Leatherwood and Dalheim, 1978 Reeves and Mitchell, 1988). Seasonal movement is associated with prey availability (e.g. Kuningas et al., 2013; Sigurjónsson and Leatherwood, 1988). Killer whales are distributed across the North Atlantic, with increasing density at higher latitudes where productivity is also higher (Forney and Wade, 2007; Hammond and Lockyer, 1988). Although killer whales were sometimes seen in the Mediterranean Sea prior to the 2000s (Table 1 and Fig. 1), there is no evidence of their continued regular occurrence within the Basin. Even with some dedicated survey effort searching for cetaceans in the Mediterranean Sea, there has been a lack of killer whale sightings since the 2000s (Table 1 and Fig. 1).

Killer whales are regularly seen in the Strait of Gibraltar, which is the natural passage between the Atlantic Ocean and the Mediterranean Sea (de Stephanis et al., 2008; Esteban et al., 2013; Guinet et al., 2007). Killer whale distribution in the southern Iberian Peninsula is related to the migration of its main prey, eastern Atlantic bluefin tuna (Esteban et al., 2013) (Fig. 2). This migratory pelagic tuna undertakes annual reproductive migrations, entering the Mediterranean Sea from the Atlantic Ocean during late spring, in search of suitable spawning areas (Sella, 1929). Esteban et al. (2013) used cetacean survey data (2002–2012, 322 killer whale sightings and 10,952 sightings of other cetaceans used as pseudo-absences of killer whales) combined with predictive generalized additive models to investigate probable factors affecting killer whale distribution in the Strait of Gibraltar, Gulf of Cadiz, and Alborán Sea. Models indicated that the killer whales were associated with the probable distribution of bluefin tuna migrating through the study area, thus restricting whale distribution to the shallow waters of the outer coastal Strait of Gibraltar and Gulf of Cadiz region in spring and to the shallow waters of the Strait of Gibraltar in spring and summer (de Stephanis et al., 2008; Esteban et al. 2013), when shoals of tuna migrate through the area (de la Serna et al., 2004).

Table 1 Sighting, Stranding, and Capture Information of Killer Whales (*Orcinus orca*) in the Mediterranean Sea and Surrounding Waters

Date (Day/Month/Year)	Latitude	Longitude	Type of Observation	References
22/06/1787	43.31	6.75	Capture	Van Beneden (1881)
1886	39.86	0.66	Sighting from land	Brú (1913)
13/12/1890	43.46	−3.78	Stranding	Casinos and Vericad (1976)
22/06/1902	36.33	−5.15	Capture	Casinos and Vericad (1976)
1914	43.46	−3.78	Stranding	Casinos and Vericad (1976)
26/12/1941	39.7	3.45	Stranding	Casinos and Vericad (1976)
1966	40.07	4.09	Stranding	Hammond and Lockyer (1988)
26/05/1970	36.56	−6.20	Stranding	Valverde (2006)
Before 1973	43.33	3.58	Stranding	Duguy and Cyrus (1973)
Before 1973	42.41	3.18	Stranding	Duguy and Cyrus (1973)
01/05/1973	40.08	0.14	Stranding	Casinos and Filella (1975)
1978–1982	39.63	10.48	Sighting at sea	Di Natale and Mangano (1983)
1978–1982	38.46	12.80	Sighting at sea	Di Natale and Mangano (1983)
1978–1982	38.98	15.57	Sighting at sea	Di Natale and Mangano (1983)
1978–1982	36.73	18.18	Sighting at sea	Di Natale and Mangano (1983)
12/09/1982	38.00	25.00	Sighting at sea	Hammond and Lockyer (1988)
09/03/1983	37.30	−0.55	Sighting at sea	Hammond and Lockyer (1988)
17/06/1983	36.11	−4.03	Sighting at sea	Hammond and Lockyer (1988)
27/06/1984	39.00	9.42	Sighting at sea	Raga et al. (1985)
27/06/1984	39.00	9.70	Sighting at sea	Hammond and Lockyer (1988)
14/08/1985	43.55	8.5	Sighting at sea	Notarbartolo di Sciara (1987)
16/08/1985	43.53	7.77	Sighting at sea	Notarbartolo di Sciara (1987)
1/10/1985	43.86	8.4	Sighting at sea	Notarbartolo di Sciara (1987)
02/02/1986	37.95	24.75	Sighting at sea	Hammond and Lockyer (1988)
07/08/1990	36.75	−3.83	Stranding	Borrell et al. (2000)

Continued

Table 1 Sighting, Stranding, and Capture Information of Killer Whales (*Orcinus orca*) in the Mediterranean Sea and Surrounding Waters—cont'd

Date (Day/Month/Year)	Latitude	Longitude	Type of Observation	References
09/12/1990	36.73	−3.58	Stranding	Borrell et al. (2000)
08/05/1991	36.50	−4.86	Stranding	Borrell et al. (2000)
05/08/1991	36.50	−4.86	Stranding	Borrell et al. (2000)
01/03/1992	39.86	39.86	Sighting from land	Belenguer and Kersting (2011)
1997	36.48	−6.25	Stranding	Gutiérrez-Expósito et al. (2012)
06/06/2004	36.95	−7.20	Sighting at sea	Baez et al. (2007)
07/06/2004	36.99	−7.12	Sighting at sea	Baez et al. (2007)
08/06/2004	36.97	−8.01	Sighting at sea	Baez et al. (2007)
10/06/2004	36.92	−8.05	Sighting at sea	Baez et al. (2007)
12/06/2004	36.93	−7.50	Sighting at sea	Baez et al. (2007)
12/04/2012	36.28	−4.34	Sighting from land	T. Todorov (Iberian Peninsula, Spain, personal communication)
14/04/2012	36.27	−4.38	Sighting from land	T. Todorov (Iberian Peninsula, Spain, personal communication)

3. THE STRAIT OF GIBRALTAR KILLER WHALES

In the Strait of Gibraltar, 47 individual killer whales were identified between 1999 and 2011 (Esteban et al., 2016a). Nine individuals were considered to have died over the study period (one female stranded in 2006, known as 'Vega'; two juveniles; five calves that were considered dead, as they were not seen with their mother or within their pod for at least three consecutive years, and another adult individual that was not observed within its pod after 2007—recognizing that this individual may have left the area). After 2005, all newly identified individuals were calves (nine individuals) from known females of existing pods (Esteban et al., 2016b).

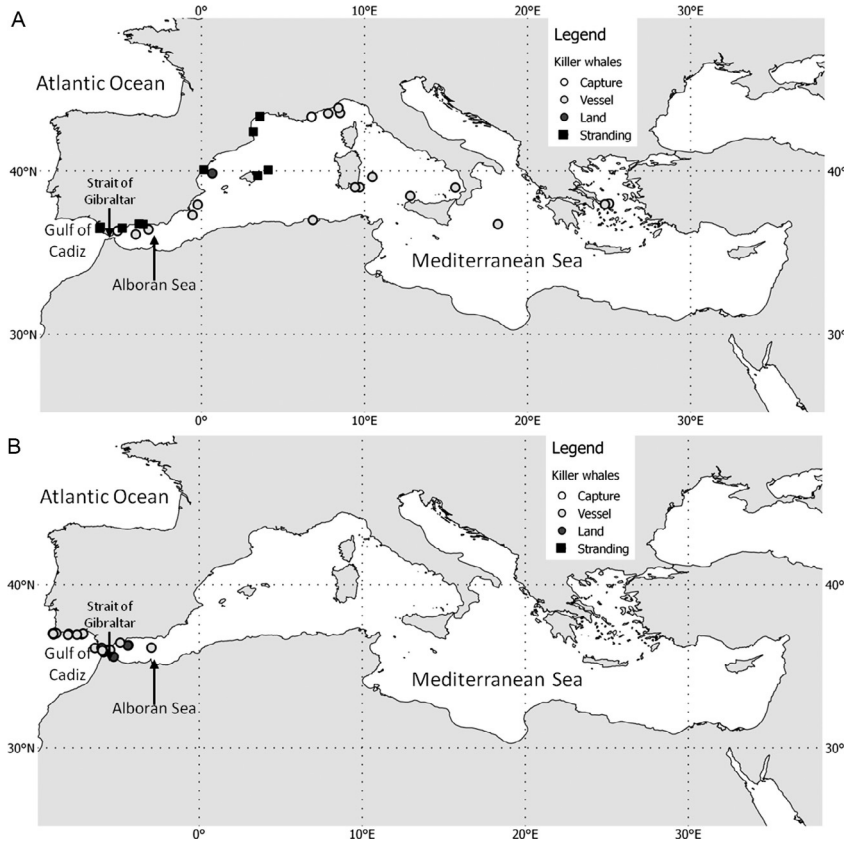


Fig. 1 Killer whale (*Orcinus orca*) distribution in the Mediterranean Sea and adjacent waters. (A) Before 2000 and (B) after 2000. *Based upon opportunistic observations and data from Esteban, R., Verborgh, P., Gauffier, P., Giménez, J., Afán, I., Cañadas, A., García, P., Murcia, J., Magalhães, S., Andreu, E., de Stephanis, R., 2013. Identifying key habitat and seasonal patterns of a critically endangered population of killer whales. J. Mar. Biol. Assoc. U.K. 94, 1317–1325. doi:10.1017/S002531541300091X.*

The first cumulative total abundance count (calculated in 2005) was 32 individuals. The count remained fairly constant after that time because most of the calves born subsequently died during this period, except for three new unidentified individuals which were added to the catalogue after 2005. In 2011, four births increased the count of that year to 39 individuals (Esteban et al., 2016a) (see also Section 6).

As has been well described for this species (e.g. Bigg et al., 1990), killer whales in the Strait of Gibraltar have a matrilineal social structure, and

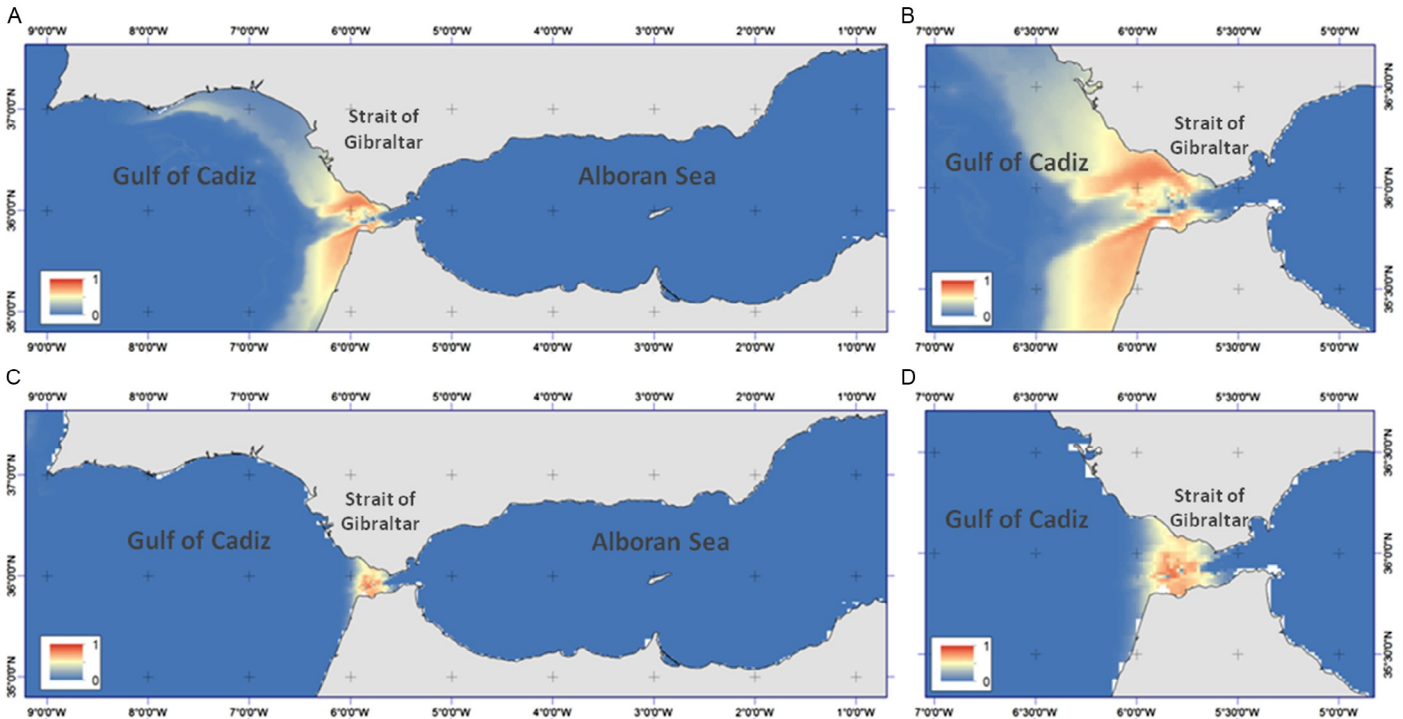


Fig. 2 Predictive density surface map of killer whales (*Orcinus orca*) based on spatial modelling: in spring, best predicted model (A) at southern Iberian Peninsula and (B) in the Strait of Gibraltar; in summer, best predicted model (C) at southern Iberian Peninsula and (D) in the Strait of Gibraltar. *Esteban, R., Verborgh, P., Gauffier, P., Giménez, J., Afán, I., Cañadas, A., García, P., Murcia, J., Magalhães, S., Andreu, E., de Stephanis, R., 2013. Identifying key habitat and seasonal patterns of a critically endangered population of killer whales. J. Mar. Biol. Assoc. U.K. 94, 1317–1325. doi:10.1017/S002531541300091X.*

certain individuals tend to have strong and enduring associations with specific individuals (Esteban et al., 2016b,c). These killer whales were originally observed, during the early 2000s, in four identifiable pods (known as A, B, C, and D pods). All individuals remained in the pod in which they were sighted for the first time. However in 2006 (due to the birth of one individual and the death of another), pod A underwent fission to form two socially differentiated pods (A1 and A2), resulting in a total of five identifiable pods (A1, A2, B, C, and D pods) (see Esteban et al. (2016b) (see also Section 5).

Of the 39 individual Strait of Gibraltar killer whales in 2011, 18 individuals were from pod A, nine from pod B, six from pod C, and six from pod D. At the start of our research, in 1999, 11 individuals were identified in pod A. Individuals from pods A1 and A2 were sighted every year from 1999 to 2011 (1999–2005 as members of pod A). Between 1999 and 2005, five viable calves were born (and one juvenile died), increasing the number of whales in pod A to 15 individuals by 2005. In 2011, three births occurred in pod A1, and one in pod B (see Esteban et al., 2016a).

4. ECOLOGY

4.1 Diet

Killer whales are apex predators, consuming a wide range of prey species, including squid, octopus, bony and cartilaginous fish, sea turtles, seabirds, sea and river otters, dugongs, pinnipeds, and cetaceans (Heyning and Dahlheim, 1988). Nevertheless, individual populations of killer whales regularly appear to specialize in particular types of prey (Ford et al., 1998).

Killer whales in the Strait of Gibraltar feed seasonally mainly on bluefin tuna (Esteban et al., 2016b). The diet of killer whales in the Strait of Gibraltar has been investigated using stable isotope analysis of skin samples (collected from May to July) (García-Tiscar, 2009). Isotope data collected from the female killer whale that stranded in 2006 ('Vega'), indicated that the animal had fed on other coastal fish species (García-Tiscar, 2009). Interestingly, this individual may have belonged to a migrant lineage, due to ancestry shared with individuals from both the Canary Islands and the Strait of Gibraltar (Esteban et al., 2016c). It is also possible that those killer whales opportunistically feed on coastal fish species in and around the Strait of Gibraltar.

4.2 Feeding Strategies

Killer whales in the Strait of Gibraltar use at least two different hunting techniques to feed on tuna. In spring, whales have been observed to chase tuna for up to 30 min at a relatively high speed (3.7 ± 0.2 m s^{-1}) until they captured them (Guinet et al., 2007). This has been described as an endurance-exhaustion technique, wherein killer whales drive tuna beyond their aerobic limits until the fish are exhausted and can be captured. Using this technique killer whales apparently capture small to medium size (0.8–1.5 m) tuna (Guinet et al., 2007). Killer whales have also been observed actively hunting in summer (Esteban et al., 2013). However, the whales were more frequently observed to be interacting with the artisanal drop-line fishery (Esteban et al., 2016a). The fishery began operating in 1995 (July–August) in the central waters of the Strait of Gibraltar (Srour, 1994). Killer whales have been interacting with the fishery since at least 1999 at the start of our observations in the Strait, and their distribution appears to be linked to the presence of drop-line boats in the area (de Stephanis et al., 2008). The whales patrol the vicinity of the boats until they find a tuna hooked on a line, and then depredate the fish before fishermen can bring it to the surface. Not all the killer whales sighted in the Strait have been observed to interact with the fishery (Esteban et al., 2013, 2016a).

4.3 Energy Requirements

Recently, Esteban et al. (2016a) investigated killer whale foraging behaviour and estimated energy requirements of 39 killer whales actively hunting and/or interacting with the drop-line fishery in the Strait of Gibraltar between 1999 and 2011. Esteban et al. (2016a) classified the whales into two categories: 'interacting individuals' observed interacting with the fishery at least once (INT; pods A1 and A2), and 'not interacting individuals' that were never observed interacting (NOT; pods B, C, and D). Energy requirements were estimated as tons of tuna needed for the entire subpopulation of 39 whales. Esteban et al. (2016a) estimated that (in 2011) the killer whales required approximately 1600 tons of tuna to meet their annual energy requirements. This estimate was converted to the number of fish needed, and to tons needed, to support killer whales in the Strait for each of the two feeding strategies (Esteban et al., 2016a). The interaction with the long-line fishery provided killer whales with larger tuna (2 m), compared to individuals that only hunted actively (tuna $\leq$ 0.8–1.5 m). Consequently INT whales needed fewer tuna to meet their energy requirements while

interacting (8 tuna per day) compared to those actively hunting (21–141 tuna per day) (Esteban et al., 2016a).

4.4 Interspecies Interactions

In the Strait of Gibraltar, long-finned pilot whales (*Globicephala melas*) have repeatedly been observed apparently 'mobbing' killer whales and causing them to move out of the area (de Stephanis et al., 2015). However, isotopic analysis indicated different feeding niches for the two species. Interestingly, satellite tracking showed no overlap in the distribution of one pilot whale and one killer whale tagged for 21 days simultaneously (de Stephanis et al., 2015). Consequently, neither resource competition nor interspecific predation could explain the mobbing behaviour. The authors suggested that a possible historical presence of mammal-eating killer whales in the area could have provoked this behaviour as an antipredator defence that was maintained over time (de Stephanis et al., 2015).

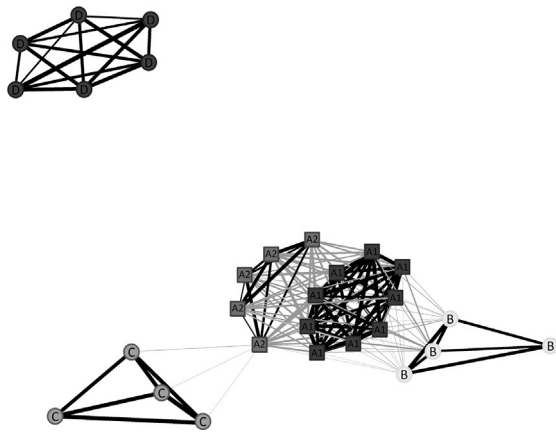
5. SOCIAL STRUCTURE

Social structure in killer whales has been widely described and investigated and is typically characterized by stable hierarchically structured social units and strong natal philopatry (e.g. Bigg et al., 1990). However, several studies have described disparity in sociality among populations of killer whales (e.g. in mammal-eating 'transient' and fish-eating 'resident' killer whales off southern Vancouver Island) (Baird and Whitehead, 2000). A recent study of killer whales in the Northeast Atlantic suggested that sociality is plastic and can be modified depending on local ecological conditions (Beck et al., 2012). Furthermore, in the Northeast Pacific, group size of killer whales was related to prey abundance and the Pacific Decadal Oscillation (Lusseau et al., 2004). Anthropogenic activities have also been proposed to influence the social structure and behaviour of killer whales. Williams and Lusseau (2006) have suggested that targeted removals of individuals for dolphinaria may have altered the social structure into isolated groups in north-eastern Pacific killer whale populations. These populations are also thought to be experiencing social fragmentation related to low abundance of their primary prey, Chinook salmon (*Oncorhynchus tshawytscha*) (Foster et al., 2012; Parsons et al., 2009). Thus, disparity in social structure within populations can be affected by anthropogenic activities and changes in local ecological conditions.

Esteban et al. (2016b) found that foraging strategies are different between pods, and all members within each pod participate in the same strategy. All individuals from A1 and A2 pods were observed actively hunting tuna as well as interacting with the drop-line fishery annually during the summer. They were also seen actively hunting in spring. Pods B, C, and D were only observed actively hunting. Pods C and D were only seen actively hunting in spring, and pod B was seen hunting in both spring and summer (Fig. 3).

Such foraging strategies could arise through a combination of vertical transmission within matrilineal pods or by vertical and horizontal transmission with individuals aligning their behaviour with that of other pod members (Boyd and Richerson, 1985). Notably, all members of A pod were interacting with the drop-line fishery as early as 1999, indicating that this behaviour spread throughout A pod in less than a generation after the fishery began in 1995. Consequently, the transmission of this new behaviour could not have occurred only vertically from mother to calves. This within-pod uniformity and between-pod differentiation in a recently derived behaviour is consistent with, but not proof of, selective transmission via social learning (e.g. Galef, 1992).

Esteban et al. (2016b) proposed that social structure in these whales is shaped by maternal kinship and foraging behaviour. In the early 2000s, only


B
B
B
B
B
A2
A2
A2

one cohesive pod (A) interacted with the drop-line fishery, but after pod A underwent fission to form two socially differentiated pods (A1 and A2), each of these pods interacted with the fishery. Hence, social structure can influence the spread of novel foraging behaviours and may drive population fragmentation in this already small population. Observations of social changes at the earliest stages of diversification in foraging behaviour and social segregation may provide insights into the processes that ultimately result in the formation of socially isolated discrete ecotypes in killer whales (Esteban et al., 2016b).

6. POPULATION STRUCTURE AND MANAGEMENT UNITS

Genetic analysis of tissue samples provided information on population structuring in the north-eastern Atlantic Ocean, where three significantly differentiated populations were identified (Foote et al., 2011). Genetic analyses of skin samples collected from 85 killer whales included genotyping with 17 microsatellite loci and the sequencing of 989 bp of the mitochondrial control region for all tissue samples and the sequencing of complete mitochondrial genomes (mitogenomes; 16,386–16,392 bp) (Foote et al., 2011; and see Morin et al., 2010). The first population included individuals sampled in Norway, Iceland, and the North Sea; the second included individuals sampled in the North Sea and west coast of Iceland; and the third included individuals from the Strait of Gibraltar (10 free-ranging individuals and one stranded individual collected between 2006 and 2010) and the Canary Islands (nine individuals). The third population, which inhabits lower latitudes and was represented by a smaller sample size, consisted of maternal lineages from three highly divergent clades (Foote et al., 2011). For all three populations, all individuals sampled within a pod, shared the same mitochondrial DNA haplotype suggesting that the pods have a matrifocal social structure.

Recently Esteban et al. (2016c) adopted a multidisciplinary approach to assess ecological differences among killer whales. The authors combined photo-identification, molecular genetic analysis of individual genotypes, and isotopic biomarkers to assess contemporary gene flow and migration between killer whales in the Strait of Gibraltar and the Canary Islands. Strait of Gibraltar killer whales ($n = 39$) appeared to be reproductively, socially, and ecologically differentiated from the individuals sampled in the Canary Islands ($n = 16$) (Esteban et al., 2016c). Indeed, none of the individuals identified in the Canary Islands had ever been sighted in the Strait of Gibraltar (Fig. 4A). Additionally, genetic analyses indicated that only about 1–9% of

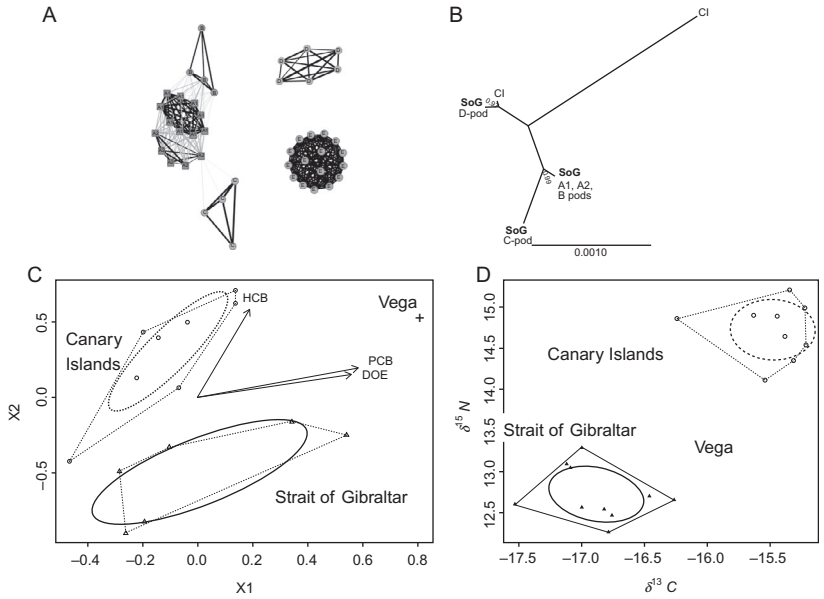


Fig. 4 Multidisciplinary study of killer whales (*Orcinus orca*) from the Strait of Gibraltar (SoG) and the Canary Islands (CI) (Esteban et al., 2016c): (A) Network diagram of killer whales including pods A1, A2, B, C, and D (including the female known as 'Vega') observed in the Strait of Gibraltar, and pod E observed in the Canary Islands. *Black edges* are within-pod associations, and *gray edges* are between-pod associations. In the SoG, *squares* indicate individuals known to interact with the drop-line fishery (INT) and *circles* indicate noninteracting (NOT) individuals. (B) Maximum-likelihood reconstruction of the genetic relationships between pods based upon complete mitochondrial genome sequences. Tip labels indicate population. Node labels indicate approximate likelihood ratio test (aLRT) support values. (C) Two-dimensional nonmetric multidimensional scaling configuration of similarities among study areas regarding 25 individual polychlorobiphenil (PCB) congeners, dichlorodiphenyldichloroethylene (*pp'*-DDE), and hexachlorobenzene (HCB) (Stress = 0.003), *dotted lines* represent convex hulls and *solid lines* the standard ellipses. (D) Stable isotopes biplot of carbon and nitrogen, *dotted lines* represent convex hulls and *solid lines* the standard ellipses. *Esteban, R., Verborgh, P., Gauffier, P., Giménez, J., Martín, V., Pérez-Gil, M., Tejedor, M., Almunia, J., Jepson, P., García-Tíscar, S., Barrettt-Lennard, L., Guinet, C., Foote, A., de Stephanis, R., 2016c. Using a multidisciplinary approach to identify a critically endangered killer whale management unit. Ecol. Indic. 66, 291–300.* http://dx.doi.org/10.1016/j.ecolind.2016.01.043.

the Strait of Gibraltar individuals were derived from the same subpopulation as Canary Islands individuals, while no complete mitogenome haplotypes were shared and no close kinship were detected between the two areas (Fig. 4B). The low genetic differentiation in combination with low levels of migration and the lack of any recent bottleneck signal are consistent with

a very recent vicariant population split (Esteban et al., 2016c). Isotopic values and pollutant loads also suggested ecological differences between study areas (Fig. 4C and D). Interestingly, the Strait of Gibraltar's pod D had never been seen in association with any of the other four pods (A1, A2, B, and C) from the Strait (see Fig. 4A). The only individual sampled from pod D was the female ('Vega') that stranded in 2006. This female was identified via genetic analysis as having a potential migrant lineage and had intermediate isotopic values and contaminant between the Strait and the Canary Islands (see Fig. 4B–D) (Esteban et al., 2016c).

Esteban et al. (2016c) concluded that based on their genetic, social, and ecological differences, killer whales from the Canary Islands and the Strait of Gibraltar should be considered as two different subpopulations and management units. Furthermore, it was noted that pod D in the Strait of Gibraltar should be considered as a third management unit, due to ecological, social, and genetic differentiation from the other Strait of Gibraltar pods (Esteban et al., 2016c).

7. DEMOGRAPHIC PARAMETERS

7.1 Abundance

The abundance of the Strait of Gibraltar killer whales (39 as of 2011) is extraordinarily low compared to most other well-studied killer whale populations (Table 2). Moreover, the low numbers of individuals found in the Strait of Gibraltar are similar to sub-Antarctic southern Indian Ocean populations that have been considered to be in decline in the Crozet Archipelago (Poncelet et al., 2010), and around Marion Island, South Africa, where inshore abundance data indicated seasonal movement offshore and/or increases in range (Reisinger et al., 2011).

7.2 Survival Rate

Esteban et al. (2016a) estimated survival rate for the two categories (INT, NOT) of killer whales in the Strait of Gibraltar, using multistate mark-recapture models (Brownie et al., 1993; Hestbeck et al., 1991). Adult survival rate for INT whales (pods A1 and A2) was estimated at 0.991 (SE: 0.014; 95% CI: 0.837–1.000) (Fig. 5). It was similar to the survival rate estimated in the northern resident population, British Columbia (Olesiuk et al., 1990, 2005), which was considered to be a stable population (Table 3).

Adult survival for NOT whales (pods B, C, and D) was estimated at 0.901 (SE: 0.067; 95% CI: 0.672–0.980) (Fig. 5). This low rate was similar

Table 2 Abundance Estimates of Killer Whale (*Orcinus orca*) Populations in Regions Worldwide from Which Estimates Are Available (See Also Forney and Wade, 2007)

Population	Abundance	References
Norway	*731	Kuningas et al. (2013)
Iceland (east coast)	*300	Beck et al. (2012) and Foote et al. (2010)
Northern Isles and northeast coast of Scotland	*50	Beck et al. (2012) and Foote et al. (2010)
West coast of Scotland, Ireland, and Wales	*10	Beck et al. (2013)
Strait of Gibraltar	*38	Esteban et al. (2016c)
North-west Atlantic	*>67	Lawson and Stevens (2013)
Eastern Arctic	*53	Young et al. (2011)
Chilean Patagonia	*49	Häussermann and Acevedo (2013)
Southern resident (North Pacific)	*88	Ellis et al. (2011)
Northern resident (North Pacific)	*260	Ellis et al. (2011)
South Alaskan resident	>700	Matkin and Durban (2011)
Aleutian Islands, Bering Sea, and Kamchatka	*, **>1500	Hoelzel et al. (2007), Ivkovich et al. (2010), Matkin and Barrett-Lennard (2007), and Matkin and Durban (2011)
West coast transient	*250	Matkin et al. (2008)
Gulf of Alaska transient	*>100	Matkin and Durban (2011)
Alaska (AT1 transient)	*7	Barrett-Lennard and Heise (2006) and Matkin et al. (2008)
New Zealand	*115	Visser (2000)
Marion Island	*37	Reisinger et al. (2011)
Crozet Archipelago	*37	Poncelet et al. (2010)

*Estimate based on photo-identification catalogues; **Estimate based on distance sampling.

to Crozet Archipelago whales (Poncelet et al., 2010), where the most reasonable explanation for the high losses was direct killing by Patagonian toothfish (*Dissostichus eleginoides*) fishers during the late 1990s early 2000s (Guinet et al., 2014) (Table 3). Juvenile survival rate could only be estimated

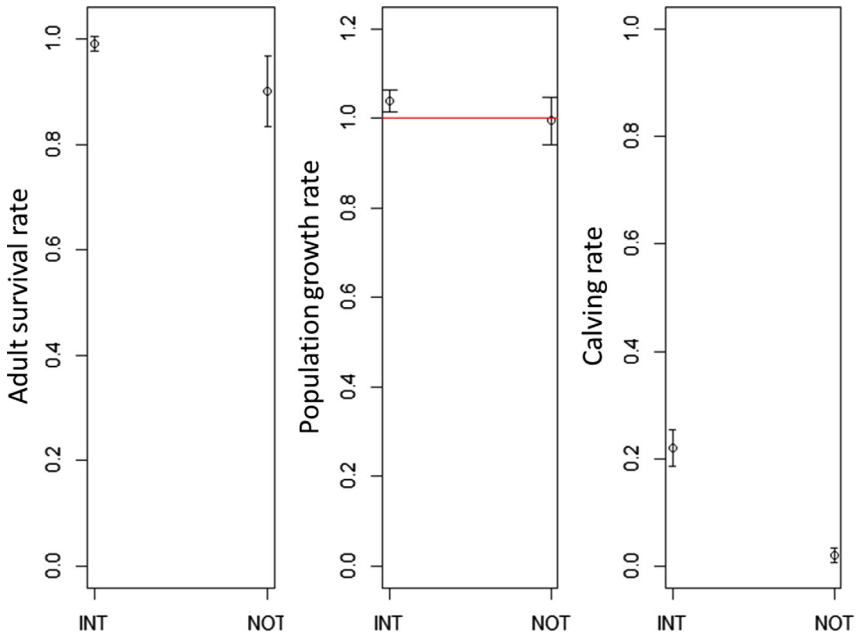


Fig. 5 Box plot demographic parameters compilation between Strait of Gibraltar killer whale (*Orcinus orca*) individuals known to interact with the drop-line fishery (INT) and noninteracting (NOT) individuals. *Red line* in population growth rate *box plot* indicates the threshold of growth of the population. *Circles* represent mean values and *lines* represent the standard error bars.

for INT whales and was 0.966 (SE: 0.031; 95% CI: 0.819–0.994). For INT whales, calf survival was one in 1999–2005 as all individuals survived their first year of life during this first period, and zero in 2006–2010, as no calf survived during this second period (Esteban et al., 2016a). Simultaneously, from 2005 to 2011 drop-line catches declined dramatically, reflecting the decline of tuna abundance in the Strait. Differences in survival rates between the two groups could be explained by the fact that depredation is a good opportunity for killer whales to feed in summer, when tuna are leaving the Strait for the Atlantic (Wilson and Block, 2009). These tuna use deep waters to cross the Strait and would likely be unavailable to killer whales (Aranda et al., 2013). However, drop-line fisheries bring tuna to the surface and could be acting as an accumulator of fish, locally increasing prey availability. This is likely especially important for killer whales when abundance of tuna is low. Food provisioning through depredation could have positively affected INT (pods A1 and A2) killer whale life history parameters in the years in which the tuna stock was at its lowest. This relationship between calf survival and tuna catches (as a proxy of tuna abundance) highlights

Table 3 Comparison of Demographic Parameters Among Killer Whale (*Orcinus orca*) Populations from the Strait of Gibraltar, North Pacific (Northern Resident), and Crozet Archipelago, Norway

	Strait of Gibraltar		**Northern Resident**		**Crozet Archipelago**		**Norway**
Parameter	1999–2011 (Esteban et al., 2016a)		1973–1995 (Olesiuk et al., 1990)	1996–2004 (Olesiuk et al., 2005)	1988–1989 (Poncelet et al., 2010)	1998–2000 (Poncelet et al., 2010)	1986–2003 (Kuningas et al., 2013)
	INT	NOT					
Adult survival rate	0.991 (SE: 0.014)	0.901 (SE: 0.067)	0.984 females (SE: 0.004)	0.971 females (SE: 0.007)	0.942 females (95% CI: 0.844–0.980)	0.901 females (95% CI: 0.742–0.966)	0.977 females (SE: 0.009)
			0.959 males (SE: 0.08)	0.909 males (SE: 0.017)	0.935 males (95% CI: 0.817–0.979)	0.895 males (95% CI: 0.746–0.961)	0.971 males (SE: 0.008)
Juvenile survival rate	0.966 (SE: 0.031)	–	0.984 (SE: 0.003)	0.946 (SE: 0.006)			0.816 (SE: 0.167)
Newborn survival rate	1	0	0.975 (SE: 0.013)	0.914 (SE: 0.033)			
Population growth rate	1.039 (SE: 0.025)	0.995 (SE: 0.053)	1.029	–	0.94 (95% CI: 0.90–0.99)		
Calving rate	0.219 (SE: 0.034)	0.020 (SE: 0.013)					
Calving interval	6–8 (mean = 7)	–	4.9 years (SE: 0.18)	5.5 years			5.06 years (SE: 0.722)
Fecundity rate	0.14	–		0.111 (SE: 0.086)	0.159 (SE: 0.14)		

the importance of artificial food provisioning in times when prey is scarce (Esteban et al., 2016a).

7.3 Other Parameters

Esteban et al. (2016a) estimated a growth rate of 1.039 (SE: 0.025; 95% CI: 0.986–1.091) for INT individuals (pods A1 and A2), which is equivalent to a 3.9% growth rate. In comparison, southern Alaska resident killer whale population growth rate was estimated to be 3.5%, and the population was thought to have reached a maximum growth rate thanks to the increasing return of their main prey, Chinook and coho (*Oncorhynchus kisutch*) salmon (Matkin et al., 2014). On the other hand, growth rate was almost stable for NOT pods (pods B, C, and D) at 0.995 (SE: 0.053; 95% CI: 0.832–1.159) (Fig. 5 and Table 3), with almost no recruitment over 12 years and lower adult survival, putting them a greater risk due to the already very low number of animals (18 individuals). Their situation could be similar to other populations currently on the verge of extinction such as the AT1 transient killer whales in Alaska, a group of seven individuals in which no recruitment had been reported since the Exxon Valdez oil spill in 1989 (Matkin et al., 2008, 2012), and to the group of nine killer whales around north-west Scotland and western Ireland, in which no calves had been reported for 19 years (Beck et al., 2013). Mean calving rate estimated for INT whales in the Strait of Gibraltar was 0.219 (SE: 0.034) and was higher than the rate of 0.020 estimated for the NOT whales (SE: 0.013) (Esteban et al., 2016a) (Fig. 5).

Only two births, from different females, were documented within the Strait of Gibraltar NOT whales (pods B, C, and D), and calving interval and fecundity rate were not estimated. Within the INT group (pods A1 and A2), 13 births were documented. However, only two intervals between viable calves of two females were observed and ranged from six to eight (mean = seven) years, producing a fecundity rate of 0.14 calves per year (Esteban et al., 2016a). These values were within the same ranges as for other populations available in the literature (Table 3).

8. CONSERVATION THREATS AND MANAGEMENT ACTIONS

8.1 Depletion of Bluefin Tuna Stock

Conservation of the Strait of Gibraltar killer whales depends on the adequate availability of tuna. Depletion of prey may force animals to spend more of their time foraging, which could lead to energetic and nutritional stress and

subsequent reduction of reproductive rates and increased mortality rates. There is evidence to suggest that northern and southern resident killer whales are already prey limited, due to natural and anthropogenic factors affecting their preferred prey (Ford et al., 2010; Ward, 2009; Williams et al., 2011).

The eastern stock of North Atlantic bluefin tuna has been exploited for centuries, but in 1960 the industrial purse-seine and longline fleets replaced traditional fisheries in many areas of the Mediterranean Sea and the Atlantic Ocean (Fromentin and Powers, 2005). A decade later, the market development of sushi and sashimi had made it a highly lucrative fishery (Fromentin and Ravier, 2005). These new factors have caused a decline of tuna since the 1960s (Taylor et al., 2011). The International Convention for the Conservation of Atlantic Tunas (ICCAT) routinely establishes quotas for tuna. It was estimated that the established quotas were exceeded by 44% between 2005 and 2011 since most of the catches were not being declared (Gagern et al., 2013; ICCAT, 2014). This caused Atlantic bluefin tuna to be designated as Endangered on the International Union for Conservation of Nature (IUCN) Red List of Threatened Species in 2011. Between 2009 and 2011, the ICCAT fully endorsed scientific committee advice and enforced a low total allowable catch (TAC) around 18,500–22,000 tons (ICCAT, 2014). As a result, the most recent assessments show signs of biomass increase (ICCAT, 2014). Assessment of tuna indicated that spawning stock biomass peaked over 300,000 tons in the late 1950s and early 1970s, then declined to about 150,000 tons until the mid-2000s and stabilized around TAC levels established by the ICCAT at the end of the 2010s.

Conservation of the eastern Atlantic stock of bluefin tuna is essential for the future of killer whales in the Strait of Gibraltar, as they have a highly specialized diet (García-Tiscar, 2009). Similarly, in the north-eastern Pacific, the fecundity of the southern resident population has been strongly related to the abundance of their main prey, and a decrease in Chinook salmon populations caused a reduced calving rate (Ward, 2009). Moreover, Ford et al. (2010) showed that increased mortality relating to decreased prey availability was a greater factor than reduced calving rate, in driving the population decline of resident killer whales there. In the Strait of Gibraltar, the amount of tuna captured by drop-line fishery (as a proxy for tuna abundance) has been related to the survival of killer whale calves (Esteban et al., 2016a). Any decrease in the abundance of tuna could put the population of killer whales in the Strait of Gibraltar at greater risk. The 1600 tons estimated as the Strait of Gibraltar killer whale energy requirement in 2011 (see Section 4.3) represents 9% of the quota (18,500 tons) established by the

ICCAT for the eastern stock Atlantic bluefin tuna. Esteban et al. (2016a) proposed the use of ecosystem based management (see Pikitch et al., 2004) to allocate a specific annual quota for killer whales based upon their energy requirements, as was previously suggested for North Eastern Pacific southern resident killer whales under the Pacific Salmon Treaty (see Williams et al., 2011). No direct injury or retaliation by drop-line fishermen has ever been observed or reported. As a short-term urgent action, it has also been proposed to transfer a higher percentage of the quota to the artisanal drop-line fishery in the Strait from the industrial purse-seine fisheries in the Mediterranean Sea, at least until the Strait of Gibraltar killer whale community shows clear signs of stability and recovery (Esteban et al., 2016a).

8.2 Maritime Traffic

The Strait of Gibraltar is an area of heavy maritime traffic, due to the increase in commercial vessels, whale watching, ferries, and sports fishing boats in recent decades. Whale watching has become a major tourist industry in many places around the world since the 1980s (Hoyt, 2001, 2002). In addition to boosting the economy of coastal communities and providing an economic reason for the conservation of marine mammals stocks, whale watching has also proved beneficial to increase public awareness of marine mammals and the environmental problems that they face (Duffus and Dearden, 1993; Lien, 2001; Tilt, 1986). Andalusia, located at the southern end of the Iberian Peninsula, along with the Canary Islands, represent the main Spanish regions where this activity is being undertaken. In the Strait of Gibraltar, and especially Tarifa, whale watching began in the mid-1990s and in a few years has become a profitable activity in the tourism sector, with a clear upward trend: from 400 visitors in 1998 to 26,228 in 2007 (Carbó-Penche et al., 2007; Martín and Urquiola, 2001; Urquiola and de Stephanis, 2000).

In 2007, the Spanish Royal Decree for the protection of cetaceans (R.D. 1727/2007) was promulgated to prevent or minimize the impact of whale watching activities for tourist, scientific, recreational, or educational purposes (or any other circumstances in which people might interact with cetaceans). This Decree defines minimum approach distance (60 m) and maximum speed (4 kn) that must be met so as not to injure or disturb the cetaceans. However, in that same year 47% of manoeuvres displayed by commercial whale watching boats around cetaceans did not meet these standards (Salazar-Sierra et al., 2008).

Several studies have linked changes in the short-term behaviour of killer whales with boats manoeuvres (Foote et al., 2004; Kruse, 1991; Williams

et al., 2002, 2006), due to the presence and activity of vessels, the sounds produced by vessels, or a combination of these factors. These reactions have been associated with a decrease in killer whale energy efficiency, whereby individuals need to spend more time foraging to capture the same amount of food than in the absence of boats (Kriete, 1994; Williams et al., 2002, 2006). However, individuals may react in different ways to boats. In some cases no response to the vessels was observed. Evasive tactics often differed between encounters with vessels (and sex of the whale), with the number of vessels and vessel proximity, activity, size, and the noise all affecting the killer whale responses (Williams et al., 2002, 2006).

In the Strait of Gibraltar, whale watching companies organize special trips to observe killer whales during July and August, when whales spend most of the time interacting with the drop-line fishery. The increased presence of vessels around the whales may affect foraging efficiency, making it harder for whales to obtain food in a noisier environment. This difficulty may have been exacerbated from 2005 to 2011, when tuna stocks were at their lowest and tuna were likely less available. In the past few years, whale watching companies in the Strait of Gibraltar have started to consider extending their activity to watching killer whales in spring, when whales are mainly actively hunting. In this area, at the north-western most part of the Strait of Gibraltar, vessel presence is low because the shallow coastal waters are outside of the main maritime traffic lanes. However, because whales are actively hunting, they may be more affected by vessel noise, which could mask detection of the tuna by the whales using passive listening and thus could cause a reduction in feeding efficiency.

Every year in spring, military exercises occur in the main distribution area of killer whales in the Strait. For example, in 2015 the North Atlantic Treaty Organization (NATO) coordinated the *Trident Juncture* exercise in that area, with 36,000 personnel of 30 different countries.

Two seasonal management areas for the Strait of Gibraltar were proposed by the Spanish Ministry of Environment to regulate possible disturbance activities such as commercial and recreational whale watching and military exercises in the main habitat of killer whales in spring and summer (Esteban et al., 2016a; Fig. 6).

8.3 Contamination

Organochlorine contaminants such as polychlorinated biphenyls (PCBs) pose a constant health threat to animals and humans. These components

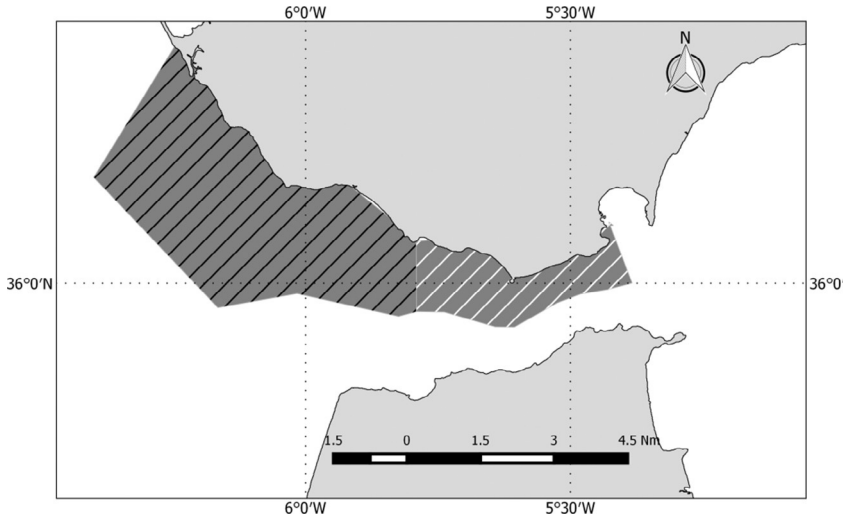


Fig. 6 Seasonal management areas proposed in the Spanish Ministry of Environment conservation plan for the killer whales (*Orcinus orca*) occurring in the Strait of Gibraltar and the Gulf of Cadiz. *Black cross lines* delimit the proposed spring management area, and *white cross lines* delimit the summer management area.

are lipophilic and persistent thus accumulate in the blubber of marine mammals. They were banned in developed countries in the 1970s and 1980s but can be still found in the blubber of animals, including marine mammals such as killer whales (e.g. Jepson et al., 2016). In cetaceans, concentrations of PCBs increase with age in males, but are reduced in females during lactation (Ross et al., 2000). Haraguchi et al. (2006) found higher concentrations of PCBs and polybrominated diphenyl ethers (PBDEs) in calves of lactating females, indicating that large amounts of halogenated organic compounds are transferred from female to offspring via lactation. In their recent meta-analysis of European data on summed 18–25 chlorobiphenyl congeners ($\sum$PCB) concentrations (mg/kg lipid weight (lw)) in odontocete cetaceans that were stranded or biopsied, Jepson et al. (2016) included tissue samples from 24 killer whales. Mean $\sum$PCB lipid concentrations in killer whales from the Northeast Atlantic were among the highest recorded in cetaceans globally. Killer whales of the Strait of Gibraltar ($n=7$ biopsied; $n=1$ stranded—the female, 'Vega') had some of the highest $\sum$PCB concentrations among cetaceans sampled (mean concentrations of 243.43 mg/kg lipid for males, and 186.74 mg/kg lipid for females of all ages sampled), markedly exceeding all known PCB toxicity thresholds for marine mammals (Jepson et al., 2016). Adult female killer whales in the Strait of Gibraltar had mean

∑PCB concentrations of 215.43 mg/kg lipid (Jepson et al., 2016). Tuna sampled in the Strait of Gibraltar also presented high levels of pollutants (Sprague et al., 2012). The transfer of organic compounds from female to calf can negatively affect calf survival, and this may have been a factor when bluefin tuna stocks were at their lowest (2005–2011), as food deprivation in the lactating females could have promoted the metabolism of lipid stores, releasing sequestered pollutants into circulation (Aguilar et al., 1999; O'Shea, 1999).

9. CONSERVATION STATUS

Killer whales in the Strait of Gibraltar were proposed as 'Critically Endangered' by the Agreement on the Conservation of Cetaceans in the Black Sea Mediterranean Sea and Contiguous Atlantic Area (ACCOBAMS) to the IUCN (Cañadas and de Stephanis, 2006), and the International Whaling Commission (IWC) recommended the implementation of a conservation plan. In 2011, the Spanish Ministry of Environment catalogued them as vulnerable (R.D. 139/2011), and a conservation plan was drafted (and is currently under revision).

Ambiguity regarding the degree of genetic isolation between Strait of Gibraltar and Canary Islands killer whales was identified as a key question in confirming their status. However, Esteban et al. (2016c) demonstrated that whales from the Strait of Gibraltar belong to a separate population that presented social, genetic, and ecological differences in comparison with the Canary Islands individuals. The IUCN defines a subpopulation as 'geographically or otherwise distinct groups in the population between which there is little demographic or genetic exchange (typically one successful migrant individual or gamete per year or less)' (IUCN, 2012). Based on this new information, it is strongly recommended that killer whales of the Strait of Gibraltar should be considered as a subpopulation for conservation purposes.

Moreover, the population of killer whales of the Strait of Gibraltar would likely be categorized as Endangered under IUCN Red List of Endangered Species criteria D (small population size), if formally assessed, because its population size is fewer than 250 mature individuals. Regional level assessments should still consider the possibility of exchange with surrounding populations. However, there are differences in behaviour, ecology, and genetics between the Strait of Gibraltar population and the Canary Islands individuals (Esteban et al. 2016c). In the Spanish Catalogue of Endangered

Species, this population of killer whales should also be considered as endangered under criteria C1 (small population size), as actual population size of the killer whale population in the Strait was 39 individuals in 2011 (Esteban et al., 2016a). This is less than 16% of the abundance of other killer whale populations considered stable or increasing in size, such as in Norway, where 731 individuals were identified (Kuningas et al., 2013), or the northern residents of the Pacific Ocean with 260 individuals (Ellis et al., 2011). In addition, based upon factors reviewed here, the Strait of Gibraltar whales should also be considered as endangered in the Spanish Catalogue of Endangered Species under criteria D (expert judgment). These factors including the major decline of killer whale primary prey, which is still overexploited, low rate of recruitment, and dependence upon catches of the artisanal dropline fishery, all endanger the Strait of Gibraltar killer whale population with the risk of extinction (Esteban et al., 2016a).

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REFERENCES

Aguilar, A., 1985. Biología y dinámica poblacional del rorcual común ("*Balaenoptera physalus*") en las aguas atlánticas ibéricas. Ph.D. dissertation, Universitat de Barcelona.

Aguilar, A., 2006. Catches of fin Whales Around the Iberian Peninsula: Statistics and Sources. International Whaling Commission, Cambridge.

Aguilar, A., Borrell, A., Pastor, T., 1999. Biological factors affecting variability of persistent pollutant levels in cetaceans. J. Cetacean Res. Manag. 1, 83–116.

Aranda, G., Abascal, F.J., Varela, J.L., Medina, A., 2013. Spawning behaviour and post-spawning migration patterns of Atlantic bluefin tuna (*Thunnus thynnus*) ascertained from satellite archival tags. PLoS One 8, e76445. http://dx.doi.org/10.1371/journal.pone.0076445.

Baez, J., Camiñas, J., Torreblanca, D., 2007. Analysis of spatial distribution for the marine bird and mammal in the Gulf of Cadiz (South-west of the Iberian Peninsula) during spring period. Bol. R. Soc. Esp. Hist. Nat. Sec. Biol. 102, 93–97.

Baird, R.W., Whitehead, H., 2000. Social organization of mammal-eating killer whales: group stability and dispersal patterns. Can. J. Zool. 78, 2096–2105. http://dx.doi.org/10.1139/cjz-78-12-2096.

Barrett-Lennard, L., Heise, K., 2006. The natural history and ecology of killer whales. In: Estes, J.A., DeMaster, D.P., Doak, D.F., Williams, T.M., Brownell, R.L. (Eds.), Whales, Whaling, and Ocean Ecosystems. University of California Press, Oakland, CA, pp. 163–173.

Beck, S., Kuningas, S., Esteban, R., Foote, A.D., 2012. The influence of ecology on sociality in the killer whale (*Orcinus orca*). Behav. Ecol. 23, 246–253. http://dx.doi.org/10.1093/beheco/arr151.

Beck, S., Foote, A.D., Kötter, S., Harries, O., Mandleberg, L., Stevick, P.T., Whooley, P., Durban, J.W., 2013. Using opportunistic photo-identifications to detect a population decline of killer whales (*Orcinus orca*) in British and Irish waters. J. Mar. Biol. Assoc. U.K. 94, 1327–1333. http://dx.doi.org/10.1017/S0025315413001124.

Belenguer, R., Kersting, D., 2011. Cetáceos en la Reserva Marina de las Islas Columbretes (Mediterráneo noroccidental): 20 años de avistamientos oportunistas. Marten Kwinkelenberg.

Bigg, M.A., Olesiuk, P.F., Ellis, G.M., Ford, J.K.B., Balcomb, K.C., 1990. Social organization and genealogy of resident killer whales (*Orcinus orca*) in the coastal waters of British Columbia and Washington State. Rep. Int. Whal. Comm. 12, 383–405.

Borrell, A., Vila, À., Forcada, J., 2000. Varamiento de cetáceos en las costas españolas del Mediterráneo durante el período 1989–1992. Miscel lània Zool. 23, 53–67.

Boyd, R., Richerson, P., 1985. Culture and the Evolutionary Process. University of Chicago Press, Chicago.

Brasseur, S., Reijnders, P., Verriopoulos, G., 1997. Mediterranean monk seal, *Monachus monachus*. In: Reijnders, P., Verriopoulos, G., Brasseur, S. (Eds.), Status of Pinnipeds Relevant to the European Union. Institute for Forestry and Nature Research, Wageningen, The Netherlands, pp. 1–195.

Brownie, C., Hines, J., Nichols, J., 1993. Capture-recapture studies for multiple strata including non-Markovian transitions. Biometrics 49, 1173–1187.

Brú, F., 1913. Notas de Caza. De José Guix, Valencia.

Cabrera, A., 1914. Fauna Ibérica; mamíferos. Museo Nacional de Ciencias Naturales, Madrid.

Cañadas, A., de Stephanis, R., 2006. Killer whale, or orca *Orcinus orca* (strait of Gibraltar subpopulation). In: Reeves, R.R., Notarbartolo di Sciara, G. (Eds.), The Status and Distribution of Cetaceans in the Black Sea and Mediterranean Sea. IUCN, Centre for Mediterranean Cooperation, Malaga, Spain, pp. 34–38.

Carbó-Penche, M., Salazar-Sierra, J.M., De Stephanis, R., Esteban, R., 2007. Socio-economic analysis of the whale-watching industry in Andalucia, Spain. In: Proceedings of the 21st Annual Meeting of the European Cetacean Society, San Sebastian, April 22–25, p. 140.

Casinos, A., Filella, S., 1975. Primer recull anual (1973) de la comissió de cetologia de la institució catalana d'Historia Natural. Bull. Inst. Catalana Hist. Nat. 39, 5–26.

Casinos, A., Vericad, J.R., 1976. The cetaceans of the Spanish coasts: a survey. Mammalia 40, 267–290.

Clapham, P., Aguilar, A., Hatch, L., 2008. Determining spatial and temporal scales for management: lessons from whaling. Mar. Mamm. Sci. 24, 183–201.

Cornide, J., 1788. Ensayo de una historia de los peces otras producciones marinas de la costa de galicia, arreglado al sistema del caballero Carlos Linneo.

Cuvier, F., 1836. De l'histoire naturelle des cétaces. Imprimerie de Félix Malteste et Cie, Paris.

De Huerta, G., 1624. Historia natural de Cayo Plinio Segundo. Traducida por el licenciado Geronimo de Huerta. Santo Oficio de la Inquisición, Médico y Familiar del.

de la Serna, J., Alot, E., Majuelos, E., Rioja, P., 2004. La migración trófica post reproductiva del atún rojo (*Thunnus thynnus*) a través del estrecho de Gibraltar. Collect. Vol. Sci. Pap. ICCAT 56, 1196–1209.

de Stephanis, R., Cornulier, T., Verborgh, P., Salazar-Sierra, J., Pérez-Gimeno, N., Guinet, C., Salazar Sierra, J., Pérez Gimeno, N., Guinet, C., 2008. Summer spatial distribution of cetaceans in the Strait of Gibraltar in relation to the oceanographic context. Mar. Ecol. Prog. Ser. 353, 275–288. http://dx.doi.org/10.3354/meps07164.

de Stephanis, R., Giménez, J., Esteban, R., Gauffier, P., García-Tiscar, S., Sinding, M.-H., Verborgh, P., 2015. Mobbing-like behavior by pilot whales towards killer whales: a response to resource competition or perceived predation risk? Acta Ethol. 18, 69–78. http://dx.doi.org/10.1007/s10211-014-0189-1.

Di Natale, A., Mangano, A., 1983. Killer whale, *Orcinus orca* (Linnaeus) and false killer whale *Pseudorca crassidens* owen, in the Italian seas. Rapp. Comm. Int. Explor. Sci. Mer. Médit. 28, 219–222.

Duffus, D., Dearden, P., 1993. Recreational use, valuation, and management, of killer whales (*Orcinus orca*) on Canada's Pacific coast. Environ. Conserv. 20, 149–156.

Duguy, R., Cyrus, J., 1973. Note préliminaire a l'étude des cétacés des côtes françaises de Méditerranée. Rev. Trav. Inst. Pêches Marit. 37, 151–158.

Duguy, R., Robineau, D., 1973. Cétacés et phoques des côtes de France. Société des sciences naturelles de la Charente-Maritime, La Rochelle, France.

Duguy, R., Casinos, A., Di Natale, A., 1983. Répartition et frequence des mammifères marins en Méditerranée. Rapp. Comm. Int. Mer. Médit. 28, 223–230.

Ellis, G., Towers, J., Ford, J., 2011. Northern Resident Killer Whales of British Columbia: Photo-Identification Catalogue and Population Status to 2010. Fisheries and Oceans Canada, Science Branch, Pacific Region.

Esteban, R., 2015. Ecology of killer whales (*Orcinus orca*) at southern Iberian Peninsula. Ph.D. dissertation, University of Cadiz, Spain.

Esteban, R., Verborgh, P., Gauffier, P., Giménez, J., Afán, I., Cañadas, A., García, P., Murcia, J., Magalhães, S., Andreu, E., de Stephanis, R., 2013. Identifying key habitat and seasonal patterns of a critically endangered population of killer whales. J. Mar. Biol. Assoc. U.K. 94, 1317–1325. http://dx.doi.org/10.1017/S002531541300091X.

Esteban, R., Verborgh, P., Gauf, P., Giménez, J., Guinet, C., de Stephanis, R., 2016a. Dynamics of killer whale, bluefin tuna and human fisheries in the Strait of Gibraltar. Biol. Conserv. 194, 31–38. http://dx.doi.org/10.1016/j.biocon.2015.11.031.

Esteban, R., Verborgh, P., Gauffier, P., Giménez, J., Foote, A., de Stephanis, R., 2016b. Maternal kinship and fisheries interaction influences killer whale social structure. Behav. Ecol. Sociobiol. 70, 111–122. http://dx.doi.org/10.1007/s00265-015-2029-3.

Esteban, R., Verborgh, P., Gauffier, P., Giménez, J., Martín, V., Pérez-Gil, M., Tejedor, M., Almunia, J., Jepson, P., García-Tíscar, S., Barrettt-Lennard, L., Guinet, C., Foote, A., de Stephanis, R., 2016c. Using a multi-disciplinary approach to identify a critically endangered killer whale management unit. Ecol. Indic. 66, 291–300. http://dx.doi.org/10.1016/j.ecolind.2016.01.043.

Foote, A.D., Osborne, R.W., Hoelzel, A.R., 2004. Environment: whale-call response to masking boat noise. Nature 428, 910. http://dx.doi.org/10.1038/428910a.

Foote, A.D., Newton, J., Piertney, S.B., Willerslev, E., Gilbert, M.T.P., 2009. Ecological, morphological and genetic divergence of sympatric North Atlantic killer whale populations. Mol. Ecol. 18, 5207–5217.

Foote, A.D., Similä, T., Víkingsson, G.a., Stevick, P.T., 2010. Movement, site fidelity and connectivity in a top marine predator, the killer whale. Evol. Ecol. 24, 803–814. http://dx.doi.org/10.1007/s10682-009-9337-x.

Foote, A.D., Vilstrup, J.T., De Stephanis, R., Verborgh, P., Abel Nielsen, S.C., Deaville, R., Kleivane, L., Martín, V., Miller, P.J.O., Oien, N., Pérez-Gil, M., Rasmussen, M., Reid, R.J., Robertson, K.M., Rogan, E., Similä, T., Tejedor, M.L., Vester, H., Víkingsson, G.A., Willerslev, E., Gilbert, M.T.P., Piertney, S.B., Nielsen, S.C.A., Øien, N., Pasmussen, M., 2011. Genetic differentiation among North Atlantic killer

whale populations. Mol. Ecol. 20, 629–641. http://dx.doi.org/10.1111/j.1365-294X.2010.04957.x.

Ford, J.K.J., Ellis, G.G.M., Barrett-Lennard, L.G., Morton, A.B., Palm, R.S., Balcomb III, K.C., 1998. Dietary specialization in two sympatric populations of killer whales (*Orcinus orca*) in coastal British Columbia and adjacent waters. Can. J. Zool. 76, 1456–1471. http://dx.doi.org/10.1139/z98-089.

Ford, J.K.B., Ellis, G.M., Olesiuk, P.F., Balcomb, K.C., 2010. Linking killer whale survival and prey abundance: food limitation in the oceans' apex predator? Biol. Lett. 6, 139–142. http://dx.doi.org/10.1098/rsbl.2009.0468.

Forney, K.A., Wade, P., 2007. Worldwide distribution and abundance of killer whales. In: - Estes, J.A., DeMaster, D.P., Doak, D.F., Williams, T.M., Brownell Jr., R.L. (Eds.), Whales, Whaling and Ocean Ecosystems. University of California Press, Berkeley, California, pp. 145–162.

Foster, E.A., Franks, D.W., Morrell, L.J., Balcomb, K.C., Parsons, K.M., van Ginneken, A., Croft, D.P., van Ginneken, A., 2012. Social network correlates of food availability in an endangered population of killer whales, *Orcinus orca*. Anim. Behav. 83, 731–736. http://dx.doi.org/10.1016/j.anbehav.2011.12.021.

Fromentin, J.J.-M., Powers, J.J.E., 2005. Atlantic bluefin tuna: population dynamics, ecology, fisheries and management. Fish Fish. 6, 281–306. http://dx.doi.org/10.1111/j.1467-2979.2005.00197.x.

Fromentin, J., Ravier, C., 2005. The East Atlantic and Mediterranean bluefin tuna stock: looking for sustainability in a context of large uncertainties and strong political pressures. Bull. Mar. Sci. 76, 353–361.

Gagern, A., van den Bergh, J., Sumaila, U.R., 2013. Trade-based estimation of bluefin tuna catches in the Eastern Atlantic and Mediterranean, 2005–2011. PLoS One 8, e69959. http://dx.doi.org/10.1371/journal.pone.0069959.

Galef, B., 1992. The question of animal culture. Hum. Nat. 3, 151–178.

García-Tiscar, S., 2009. Interacciones entre delfines mulares y orcas con pesqeuerías en el Mar de Alborán y Estrecho de Gibraltar. Ph.D. dissertation, Universidad Autónoma de Madrid.

Graells, M., 1889. Las ballenas en las costas oceánicas de españa. Noticias recogidas e investifaciones hechas por el Dr. M.P. Graelss. Imprenta de Don Luis Aguado, Madrid, Spain.

Graus, E., Filella, S., Raga, J.A., Raduán, A., 1986. Cetáceos varados en las costas del Mediterráneo Ibérico, durante los años 1980–1981. Miscel lània Zool. 10, 353–358.

Guinet, C., Domenici, P., de Stephanis, R., Barrett-Lennard, L., Ford, J.K.B., Verborgh, P., 2007. Killer whale predation on bluefin tuna: exploring the hypothesis of the endurance-exhaustion technique. Mar. Ecol. Prog. Ser. 347, 111–119. http://dx.doi.org/10.3354/meps07035.

Guinet, C., Tixier, P., Gasco, N., 2014. Long-term studies of Crozet Island killer whales are fundamental to understanding the economic and demographic consequences of their depredation behaviour on the Patagonian toothfish fishery. ICES J. Mar. Sci. 72, 1587–1597.

Gutiérrez-Expósito, C., Rivilla, J.C., Alís, S., Máñez, M., Garrido, H., Jiménez, F.J., Cobo, M.D., 2012. Veinticinco años (1986–2011) de monitorización de varamientos de mamíferos marinos en el litoral de Doñana (Huelva, SO España). Galemys: Span. J. Mamm. 24, 1–5. http://dx.doi.org/10.7325/Galemys.2012.N09.

Hammond, P.S., Lockyer, C., 1988. Distribution of killer whales in the eastern North Atlantic. Rit. Fiskid. 11, 24–41.

Haraguchi, K., Hisamichi, Y., Endo, T., 2006. Bioaccumulation of naturally occurring mixed halogenated dimethylbipyrroles in whale and dolphin products on the Japanese market. Arch. Environ. Contam. Toxicol. 51, 135–141. http://dx.doi.org/10.1007/s00244-005-1140-2.

Häussermann, V., Acevedo, J., 2013. Killer whales in Chilean Patagonia: additional sightings, behavioural observations, and individual identifications. Rev. Biol. Mar. Oceanogr. 48, 73–85. http://dx.doi.org/10.4067/S0718-19572013000100007.

Hestbeck, J., Nichols, J., Malecki, R., 1991. Estimates of movement and site fidelity using mark-resight data of wintering Canada geese. Ecology 72, 523–533.

Heyning, J.E., Dahlheim, M.E., 1988. Orcinus orca. Mamm. Species 304, 1–9.

Hoelzel, A.R., Hey, J., Dahlheim, M.E.M., Nicholson, C., Burkanov, V., Black, N., 2007. Evolution of population structure in a highly social top predator, the killer whale. Mol. Biol. Evol. 24, 1407–1415. http://dx.doi.org/10.1093/molbev/msm063.

Horozco, A., 1598. Historia de la ciudad de Cádiz. El Excmo. Ayuntamiento de esta M.N. M. L. y M. H. Ciudad.

Hoyt, E., 2001. Whale Watching 2001: Worldwide Tourism Numbers, Expenditures, and Expanding Socioeconomic Benefits. International Fund for Animal Welfare, Yarmouth Port, MA.

Hoyt, E., 2002. Whale watching. In: Perrin, W.F., Wursig, B., 'Hans' Thewissen, J.G.M. (Eds.), Encyclopedia of Marine Mammals. Academic Press, San Diego, California, pp. 1305–1310.

ICCAT, 2014. Report of the 2014 Atlantic Bluefin Tuna Stock Assessment Session. ICCAT, Madrid, Spain.

Israëls, L., 1992. Thirty Years of Mediterranean Monk Seal Protection, a Review. Instituut voor Taxonomische Zoölogie/Zoölogische Museum, Universiteit van Amsterdam, Amsterdam, The Netherlands.

IUCN, 2012. Red List Categories and Criteria: Version 3.1, second ed. IUCN, Gland, Switzerland and Cambridge, UK.

Ivkovich, T., Filatova, O.a., Burdin, A.M., Sato, H., Hoyt, E., 2010. The social organization of resident-type killer whales (*Orcinus orca*) in Avacha Gulf, Northwest Pacific, as revealed through association patterns and acoustic similarity. Mamm. Biol.—Z. Säugetierkd. 75, 198–210. http://dx.doi.org/10.1016/j.mambio.2009.03.006.

Jepson, P., Deaville, R., Barber, J., Aguilar, A., Borrell, A., Murphy, S., Barry, J., Brownlow, A., Barnett, J., Berrow, S., Cunningham, A., Davison, N., ten Doeschate, M., Esteban, R., Ferreira, M., Foote, A., Genov, T., Giménez, J., Loveridge, J., Llavona, A., Martin, V., Maxwell, D., Papachlimitzou, A., Penrose, R., Perkins, M., Smith, B., de Stephanis, R., Tregenza, N., Verborgh, P., Fernandez, A., Law, R., 2016. PCB pollution continues to impact populations of orcas and other dolphins in European waters. Sci. Rep. 6, 18573. http://dx.doi.org/10.1038/srep18573.

King, J., 1956. The monk seals (genus Monachus). Bull. Br. Mus. Zool. Lond. 3, 203–256.

Kriete, B., 1994. Bioenergetics in the Killer Whale, *Orcinus orca*. Ph.D. dissertation, University of British Columbia, British Columbia, Vancouver, Canada.

Kruse, S., 1991. The interactions between killer whales and boats in Johnstone Strait, BC. In: Pryor, K., Norris, K.S. (Eds.), Dolphin Societies: Discoveries and Puzzles. University of California Press, Berkeley, California, pp. 148–159.

Kuningas, S., Similä, T., Hammond, P.S., 2013. Population size, survival and reproductive rates of northern Norwegian killer whales (*Orcinus orca*) in 1986–2003. J. Mar. Biol. Assoc. U.K. 94, 1277–1291. http://dx.doi.org/10.1017/S0025315413000933.

Lawson, J.W., Stevens, T.S., 2013. Historic and current distribution patterns, and minimum abundance of killer whales (*Orcinus orca*) in the north-west Atlantic. J. Mar. Biol. Assoc. U.K. 94, 1253–1265. http://dx.doi.org/10.1017/S0025315413001409.

Leatherwood, J.S., Dalheim, M.E., 1978. Worldwide distribution of pilot whales and killer whales. Nav. Ocean Syst. Cent. Tech. Rep. 443, 1–39.

Lien, J., 2001. The conservation basis for the regulation of whale watching in Canada by the Department of Fisheries and Oceans: a precautionary approach.

Lusseau, D., Williams, R., Wilson, B., Grellier, K., Barton, T., Hammond, P., Thompson, P., 2004. Parallel influence of climate on the behaviour of Pacific killer whales and Atlantic bottlenose dolphins. Ecol. Lett. 7, 1068–1076. http://dx.doi.org/10.1111/j.1461-0248.2004.00669.x.

Machado y Nuñez, A., 1869. Catálogo metódico y razonado de los mamíferos de andalucia, clasificados segun el sistema del Dr. Enrique Schinz. Imprenta de Gironés y Orduña, Sevilla, Spain.

Martín, V., Urquiola, E., 2001. La observación de cetáceos. In: Naturaleza de Las Islas Canarias. Ecología Y Conservación. Turquesa, Spain, Santa Cruz de Tenerife, pp. 289–295.

Matkin, C., Barrett-Lennard, L., 2007. Ecotypic variation and predatory behavior among killer whales (*Orcinus orca*) off the eastern Aleutian Islands, Alaska. Fish. Bull. 105, 74–87.

Matkin, C., Durban, J., 2011. Killer whales in Alaskan waters. J. Am. Cetacean Soc. 40, 24–29.

Matkin, C., Saulitis, E., Ellis, G., Olesiuk, P., Rice, S.D.S., 2008. Ongoing population-level impacts on killer whales *Orcinus orca* following the 'Exxon Valdez' oil spill in Prince William Sound, Alaska. Mar. Ecol. Prog. Ser. 356, 269–281. http://dx.doi.org/10.3354/meps07273.

Matkin, C.O., Durban, J.W., Saulitis, E.L., Andrews, R.D., Straley, J.M., Matkin, D.R., Ellis, G.M., 2012. Contrasting abundance and residency patterns of two sympatric populations of transient killer whales (*Orcinus orca*) in the northern Gulf of Alaska. Fish. Bull. 110, 143–155.

Matkin, C.O., Testa, J.W., Ellis, G.M., Saulitis, E.L., 2014. Life history and population dynamics of southern Alaska resident killer whales (*Orcinus orca*). Mar. Mamm. Sci. 30, 460–479.

Morin, P.A., Archer, F.I., Foote, A.D., Vilstrup, J., Allen, E.E., Wade, P., Durban, J., Parsons, K., Pitman, R., Li, L., Bouffard, P., Abel Nielsen, S.C., Rasmussen, M., Willerslev, E., Gilbert, M.T.P., Harkins, T., 2010. Complete mitochondrial genome phylogeographic analysis of killer whales (*Orcinus orca*) indicates multiple species. Genome Res. 20, 908–916. http://dx.doi.org/10.1101/gr.102954.109.

Notarbartolo di Sciara, G., 1987. Killer whale, *Orcinus orca*, in the Mediterranean Sea. Mar. Mamm. Sci. 3, 356–360.

O'Shea, T., 1999. Environmental contaminants and marine mammals. Sci. Total Environ. 154, 141–151.

Olesiuk, P.F., Bigg, M.A., Ellis, G., 1990. Life history and population dynamics of northern resident killer whales (*Orcinus orca*) in British Columbia. Rep. Int. Whal. Comm. (Special Issue 12), 209–244.

Olesiuk, P.F., Ellis, G.M., Ford, J.K.B., 2005. Life History and Population Dynamics of Northern Resident Killer Whales (*Orcinus orca*) in British Columbia. Fisheries and Oceans Canada, Nanaimo, BC.

Parsons, K.M.M., Balcomb, K.C.C.I., Ford, J.K.B., Durban, J.W., 2009. The social dynamics of southern resident killer whales and conservation implications for this endangered population. Anim. Behav. 77, 963–971. http://dx.doi.org/10.1016/j.anbehav.2009.01.018.

Pikitch, E., Santora, C., Babcock, E., 2004. Ecosystem-based fishery management. Science 305, 346–347.

Poncelet, E., Barbraud, C., Guinet, C., 2010. Population dynamics of killer whales (*Orcinus orca*) in Crozet Archipelago, southern Indian Ocean: a mark-recapture study from 1977 to 2002. J. Cetacean Res. Manag. 11, 41–48.

Raga, J.A., Raduán, A., Blanco, C., 1985. Contribución al estudio de la distribución de cetàceos en el Mediterráneo y Atlantico Ibérico. Miscel lània Zool. 9, 361–366.

Reeves, R.R., Mitchell, E., 1988. Killer whale sightings and takes by American pelagic whalers in the North Atlantic. Rit. Fiskid. 11, 7–23.

Reisinger, R., de Bruyn, P., Bester, M., 2011. Abundance estimates of killer whales at sub-antarctic Marion Island. Aquat. Biol. 12, 177–185. http://dx.doi.org/10.3354/ab00340.
Ronald, K., Healey, P., 1978. The monk seal (*Monachus monachus*). In: FAO Advisory and Committee on Marine Resources Research (Ed.), Mammals in the Seas. Food and Agriculture Organization of the United Nations, Rome. ISBN: 92-5-100514-1, p. 534.
Ross, P.S., Ellis, G.M., Ikonomou, M.G., Barrett-Lennard, L.G., Addison, R.F., 2000. High PCB concentrations in free-ranging Pacific killer whales, *Orcinus orca*: effects of age, sex and dietary preference. Mar. Pollut. Bull. 40, 504–515. http://dx.doi.org/10.1016/S0025-326X(99)00233-7.
Sal'nikov, M., 1959. Novi dani pro tyulenya-monakha v Chernomu Mori. Nauch. Zap. Odes. Biol., St. AN URSR.
Salazar-Sierra, J.M., Torres Barranco, F.J., Cabaleiro Mora, E., 2008. Estudio de los aspectos socioeconómicos y de conservación de la actividad de avistamiento de cetáceos. Fundación Migres, Algeciras, Spain.
Sanpera, C., Aguilar, A., 1992. Modern whaling off the Iberian Peninsula during the 20th century. Rep. Int. Whal. Comm. 42, 723–730.
Sella, M., 1929. Biologia e pesca del tonno (*Thunnus thynnus* L.). Atti Conv. Biol. Mar. Appl. Pesca, Messina, Italy, June 1928, pp. 1–32.
Sigurjónsson, J., Leatherwood, S., 1988. The Icelandic live-capture fishery for killer whales, 1976–1988. Rit. Fiskid. 11.
Sprague, M., Dick, J., Medina, A., Tocher, D., 2012. Lipid and fatty acid composition, and persistent organic pollutant levels in tissues of migrating Atlantic bluefin tuna (*Thunnus thynnus*, L.) broodstock. Environ. Pollut. 171, 61–71.
Srour, A., 1994. Développement de la nouvelle pêcherie artisanale au thon rouge dans la région de Ksar sghir. Note d'information ISPM 26, 10–11.
Taylor, N.G.N., McAllister, M.M.K., Lawson, G.G.L., Carruthers, T., Block, B.A., 2011. Atlantic bluefin tuna: a novel multistock spatial model for assessing population biomass. PLoS One 6, e27693. http://dx.doi.org/10.1371/journal.pone.0027693.
Tilt, W., 1986. Whale Watching in California: Survey of Knowledge and Attitudes. Yale School of Forestry and Environmental Studies, Yale University, New Haven, Connecticut.
Urquiola, R., de Stephanis, R., 2000. Growth of whale watching in Spain. The success of the platforms in south mainland, new rules. In: Evans, P., Pitt-Aiken, R., Rogan, R. (Eds.), 14th Annual Conference of the European Cetacean Society, Cork, Ireland, pp. 198–204.
Valverde, J., 2006. Memorias de un Biólogo heterodoxo. Pesca, ballenas, barcas. Reflexiones y cuaderno de dibujos. Quercus, Madrid, Spain.
Van Beneden, M., 1881. Mémoire sur les orques observés dans les Mers D'Europe. L'Académie, Bruselles.
Van Beneden, M., 1888. Histoire naturelle des delphinides des mers d'Europe. In: Mémoires Couronnés et Autres Mémoires. l'Académie royale de Belgique, Bruselles, p. 256.
Visser, I., 2000. Orca (*Orcinus orca*) in New Zealand waters. Ph.D. dissertation, University of Auckland, Auckland, New Zealand.
Ward, E., 2009. Quantifying the effects of prey abundance on killer whale reproduction. J. Appl. Ecol. 46, 632–640.
Williams, R., Lusseau, D., 2006. A killer whale social network is vulnerable to targeted removals. Biol. Lett. 2, 497–500. http://dx.doi.org/10.1098/rsbl.2006.0510.
Williams, R., Bain, D., Ford, J.K.B., Trites, A.W., 2002. Behavioural responses of male killer whales to a leapfrogging vessel. J. Cetacean Res. Manag. 4, 305–310.
Williams, R., Lusseau, D., Hammond, P.S., 2006. Estimating relative energetic costs of human disturbance to killer whales (*Orcinus orca*). Biol. Conserv. 133, 301–311. http://dx.doi.org/10.1016/j.biocon.2006.06.010.

Williams, R., Krkošek, M., Ashe, E., Branch, T.A., Clark, S., Hammond, P.S., Hoyt, E., Noren, D.P., Rosen, D., Winship, A., 2011. Competing conservation objectives for predators and prey: estimating killer whale prey requirements for Chinook salmon. PLoS One 6, e26738. http://dx.doi.org/10.1371/journal.pone.0026738.

Wilson, S., Block, B., 2009. Habitat use in Atlantic bluefin tuna *Thunnus thynnus* inferred from diving behavior. Endanger. Species Res. 10, 355–367. http://dx.doi.org/10.3354/esr00240.

Young, B.G., Higdon, J.W., Ferguson, S.H., 2011. Killer whale (*Orcinus orca*) photo-identification in the eastern Canadian Arctic. Polar Res. 30, 1–7. http://dx.doi.org/10.3402/polar.v30i0.7203.